Nicholas Zolman

PhD Candidate · Mechanical Engineering

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Research Statement

An interdisciplinary researcher working at the intersection of scientific machine learning, dynamics, control theory, and interpretable low-order modeling with a commitment to building community in open, inclusive, and collaborative environments.

Education

University of Washington — *Seattle, WA*

PhD, Mechanical Engineering — *2022–2027 (expected)*

- Advisor: Professor Steven L. Brunton
- Topic: Parsimonious Priors for Planning, Projections, and Physics
- MS awarded 2024

California Institute of Technology — *Pasadena, CA*

BS, Mathematics — *2012–2016*

- Graduated with Honors, emphasis on physics.

Professional Experience

2022– **Graduate Research Assistant**, University of Washington

2018–2025 **Data Scientist**, The Aerospace Corporation

- Proposed and awarded research funding for various projects: e.g. physics-informed ML, digital twins development, anomaly detection, and DRL for satellite operations.
- Led interdisciplinary teams of engineers and subject matter experts.
- Presented findings to leadership and government stakeholders.

2017–2018 **Mathematics Graduate Teaching Assistant**, UCLA

2017 **Data Science Intern**, The Aerospace Corporation

2016–2017 **Residential Counselor**, Gatton Academy of Mathematics and Science

- Lived in a residential hall on Western Kentucky University's campus with motivated STEM high school students while they took WKU classes.
- Served as a form of emotional support for students as they endured this challenging environment, while also acting as an academic support by tutoring physics, computer science, and math classes.

2016 **Undergraduate Research Assistant**, Economics, California Institute of Technology

- Worked with Professor Antonio Rangel on Bayesian methods for identifying optimal student curricula.

2015 **Summer Undergraduate Research Fellow**, Mathematics, California Institute of Technology

- Worked with Professor Matilde Marcolli on geometric methods for supersymmetry algebras.

2014 **Summer Undergraduate Research Fellow**, Astrophysics, California Institute of Technology

- Worked with Professor Philip Hopkins on analyzing galaxy mass-metallically relation from large suite of galaxy simulations.

2013 **Summer Undergraduate Research Assistant**, Physics, Western Kentucky University

- Worked with Professor Keith Andrew on modified gravity theories.

Publications

Published

Brunton, S. L., **Zolman**, **N.**, Kutz, J. N., & Fasel, U. (2025). Machine learning for sparse nonlinear modeling and control. *Annual Review of Control, Robotics, and Autonomous Systems*, *8*(1), 127–152.

Otto, S. E., **Zolman**, **N.**, Kutz, J. N., & Brunton, S. L. (2025). A unified framework to enforce, discover, and promote symmetry in machine learning. *Journal of Machine Learning Research*, *26*(248), 1–83.

Zolman, **N.**, Lagemann, C., Fasel, U., Kutz, J. N., & Brunton, S. L. (2025). SINDy-RL for interpretable and efficient model-based reinforcement learning. *Nature Communications*, *16*(1), 10714.

Mason, J. J., Allen-Blanchette, C., **Zolman**, **N.**, Davison, E., & Leonard, N. E. (2023). Learning to predict 3d rotational dynamics from images of a rigid body with unknown mass distribution. *Aerospace*, *10*(11), 921.

Marcolli, M., & **Zolman**, **N.** (2019). Adinkras, dessins, origami, and supersymmetry spectral triples. *p-Adic Numbers, Ultrametric Analysis and Applications*, *11*(3), 223–247.

Ma, X., Hopkins, P. F., Faucher-Giguère, C.-A., **Zolman**, **N.**, Muratov, A. L., Kereš, D., & Quataert, E. (2016). The origin and evolution of the galaxy mass–metallicity relation. *Monthly Notices of the Royal Astronomical Society*, *456*(2), 2140–2156.

Preprints

Howard, A. A., **Zolman**, **N.**, Jacob, B., Brunton, S. L., & Stinis, P. (2026). SINDy-KANs: Sparse identification of non-linear dynamics through Kolmogorov-Arnold networks. *arXiv preprint arXiv:2603.18548*.

Lagemann, C., Mokbel, S., Gondrum, M., Rüttgers, M., Callaham, J., Paehler, L., Ahnert, S., **Zolman**, **N.**, Lagemann, K., Adams, N., et al. (2025). Hydrogym: A reinforcement learning platform for fluid dynamics. *arXiv preprint arXiv:2512.17534*.

Andrew, K., Steinfelds, E., & **Zolman**, **N.** (2016). The Mattig expression for a d-dimensional Gauss Bonnet FRWL cosmology. *arXiv preprint arXiv:1601.07037*.

In Prep

Zolman, **N.**, Mokbel, S., Otto, S. E., & Brunton, S. L. (2026). Characterizing extreme events through sensitivity-balanced projections.

Awards, Fellowships, & Grants

2022 **Aerospace Graduate Fellowship Program**, The Aerospace Corporation

- Up to $15k for tuition towards graduate studies per year. Renewal up to 5 years.

2016 **Mari P. Ligocki Award**, California Institute of Technology

- "Given to a student who has improved the quality of student life at Caltech through their personal character."
- Awarded by the Deans Office for contributions to the Upperclassman Counselor (UCC) program, role on the Title IX Advisory Board, and annual involvement in freshman orientation.

Presentations

Invited Talks

January 2026: *SINDy-RL: Interpretable and Efficient Model-Based Reinforcement Learning*. Invited talk (virtual). The Boeing Company. Everett, WA, USA.

September 2025: *Symmetry Enforcement, Discovery, and Promotion for ML Applications in Engineering*. Invited talk. Harvard University Geometric Machine Learning Group. Cambridge, MA, USA.

September 2025: *Symmetry Enforcement, Discovery, and Promotion for ML Applications in Engineering*. Invited talk. MIT Atomic Architects Group. Cambridge, MA, USA.

March 2025: *SINDy-RL: Interpretable and Efficient Model-Based Reinforcement Learning*. Invited talk (virtual). MIT Mobility Lab. Cambridge, MA, USA.

October 2024: *SINDy-RL: Interpretable and Efficient Model-Based Reinforcement Learning*. Invited talk (virtual). University of Sydney Brain and Mind Centre. Sydney, Australia.

February 2024: *SINDy-RL: Interpretable and Efficient Model-Based Reinforcement Learning*. Invited talk. Imperial College London. Department of Aeronautics. London, United Kingdom.

February 2024. *SINDy-RL: Interpretable and Efficient Model-Based Reinforcement Learning*. Invited talk. University of Cambridge Applied and Computational Analysis series. Cambridge, United Kingdom.

Contributed Presentations

March 2026: *SINDy-RL: Interpretable and Efficient Reinforcement Learning*. Oral Presentation. ERCOFTAC ML4Fluids Workshop. Amsterdam, Netherlands.

November 2025: *Characterizing Extreme Events in Turbulent Flows through Sensitivity-Based Modal Decomposition*. Oral Presentation. APS DFD. Houston, TX, USA.

November 2025: *Characterizing Extreme Events in Turbulent Flows through Sensitivity-Based Modal Decomposition*. Oral Presentation. UW Computational Neuroscience Center Minisymposium on State Space Models and Dynamical Systems. Seattle, WA, USA.

May 2025: *Symmetry Enforcement, Discovery, and Promotion for Machine Learning Applications in Engineering*. Oral Presentation. SIAM Applications of Dynamical Systems Minisymposium: Data-Driven Modeling and Estimation for Complex Systems. Denver, CO, USA.

March 2025: *Data-Driven Reduced Order Modeling of Extreme Events in Turbulent Flows*. Oral Presentation. CFC Minisymposium: Physics-based and Data-driven Low-order Modeling for Turbulent Flows. Santiago, Chile.

February 2025: *Data-Driven Dimension Reduction Through Symmetry-Promoting Regularization*. Oral Presentation. DTE-AICOMAS Minisymposium: Deep Learning-based Reduced Order Models in Scientific Computing. Paris, France.

January 2025: *SINDy-RL: Interpretable and Efficient Reinforcement Learning*. Oral Presentation and Panelist. Joint Mathematics Meeting Minisymposium: Scientific Machine Learning: Recent Advances and Future Directions. Seattle, WA, USA.

November 2024: *Data-Driven Dimension Reduction Through Symmetry-Promoting Regularization*. Oral Presentation. APS DFD. Salt Lake City, UT, USA.

July 2024: *SINDy-RL: Interpretable and Efficient Reinforcement Learning*. Oral Presentation. WCCM Minisyposium: Synergistic computational mechanics + machine learning for the digital twinning of intelligent vehicles. Vancouver, BC, Canada.

May 2024: *SINDy-RL: Interpretable and Efficient Reinforcement Learning*. Oral Presentation. ECCOMAS Minisyposium: Recent Advances in Deep RL of Complex Dynamical Systems. Lisbon, Portugal.

February 2024: *SINDy-RL: Uncertainty-aware active learning for simultaneous dynamics discovery and control*. Oral Presentation. SIAM-UQ Minisymposium: Reduced order modeling, Learning, UQ, and their interaction. Trieste, Italy.

November 2023: *SINDy-RL: Interpretable and efficient reinforcement learning for fluid flow control*. Oral Presentation. APS DFD. Washington D.C., USA.

June 2023: *Reinforcement Learning Tutorial*. Oral Presentation. NSF AI Institute in Dynamic Systems: Machine Learning Workshop. Seattle, WA, USA.

July 2019: *Application of reinforcement learning to rendezvous and proximity operations*. Oral Presentation and Panelist. Seventh International Conference on Space Mission Challenges for Information Technology Mini-Workshop: Machine Learning for Spaced-Based Applications. Pasadena, CA, USA.

Teaching Experience

2025 (SU)	**Youth Camp: Introduction to Math Modeling**, Guest Lecturer	*Univ. of Washington*
2025 (SP)	**ENGR 521: Physics-Informed Machine Learning**, Guest Lecturer	*Univ. of Washington*
2018 (SP)	**Math 134: Introduction to Dynamical Systems**, Teaching Assistant	*UCLA*
2018 (WI)	**Math 132: Complex Analysis for Applications**, Teaching Assistant	*UCLA*
2017 (AU)	**Math 33B: Differential Equations**, Teaching Assistant	*UCLA*

Mentoring

2025– **Anuj Krovvidi**, Undergraduate Research Assistant, University of Washington *Seattle, WA*
- Project: Direct policy optimization with SINDy surrogate models.

2022 **Eric Ma**, Undergraduate Intern, The Aerospace Corporation *El Segundo, CA*
- Project: Dynamics extraction from neuromorphic sensor data.

2021 **Noah Haig**, Undergraduate Intern, The Aerospace Corporation *El Segundo, CA*
- Project: Physics-informed neural networks for anomaly detection.

2021 **Austin Marstaller**, Graduate Intern, The Aerospace Corporation *El Segundo, CA*
- Project: Physics-informed neural networks for anomaly detection.

2020–2022 **Justice Mason**, Graduate Intern, The Aerospace Corporation *El Segundo, CA*
- Project: Learning to predict 3D rotational dynamics from images of a rigid body.

2020 **Jalani Williams**, Graduate Intern, The Aerospace Corporation *El Segundo, CA*
- Project: Hamiltonian neural networks for rotating bodies.

2020 **Gaurav Kuppa**, Undergraduate Intern, The Aerospace Corporation *El Segundo, CA*
- Project: Deep reinforcement learning for robotic proximity operations.

2020 **Dhruv Patel**, Graduate Intern, The Aerospace Corporation *El Segundo, CA*
- Project: Topological data analysis for time series anomaly detection.

Outreach & Professional Development

Service and Outreach

2022– **UW Women of Aerospace**, Mentor
- 2024-25 Mentee: Neha Yelve
- 2022-24 Mentee: Jerusalem Sintayehu

2020–2022 **Aerospace Corporation Physics + ML Journal Club**, Founder and coordinator

2019–2022 **Aerospace Corporation Intern Program**, Mentor, panelist, and coordinator.

2014–2016 **Caltech Title IX Advisory Board**, House Lead

2013–2015 **Caltech Upperclassman Counselor**, Head UCC

Organization and Leadership

2026 **WCCM-ECCOMAS 2026**, Minisymposium Organizer and Session Chair *Munich, Germany*
- MS: *Cents & Sensitivities: Exploiting Differentiable Solvers for Low-Cost Optimization in Science and Engineering*

2025 **DTE-AICOMAS 2025**, Session Chair *Paris, France*
- MS: *Advancements of Data-Driven Methods in Computational Mechanics*

2025– **NSF AI Institute in Dynamic Systems**, Seminar organizer and host *Seattle, WA*
- Hybrid Seminar Series: *Data-Driven Science and Engineering*
- Identified and invited speakers from a variety of academic backgrounds
- Hosted weekly seminars, moderated in-person and virtual questions.
- Organized in-person research meetings, lunch with students, and dinner with faculty.

2023– **Data-Driven PhD "Stand-up"**, Founder and organizer *Seattle, WA*
- Led weekly meeting for graduate students across different groups focused on using ML in engineering.
- Meetings focused setting intentional goals, troubleshooting problems, and fostering community.

Peer Review

Proceedings of the National Academy of Sciences
Physical Review Letters Fluids
Learning for Dynamics and Control Conference (L4DC)

Professional Memberships

Society of Industrial and Applied Mathematicians
American Physical Society